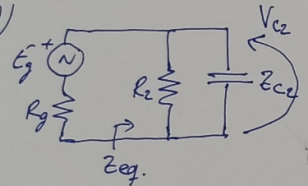
P1

a) Resolvemos el circuito antes de la conmutación, para $t<0$, en el cual se encuentra conectado el generador $e_g(t) = 4\cos(10^3 t)\,V$ desde un tiempo $t=-\infty$, esto es, el circuito está en régimen permanente sinusoidal y resolvemos usando fasores e impedancias

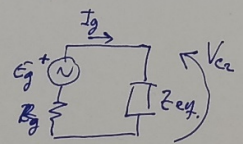
$t<0$)



$\bar{e}_g = -4\,V$, $\omega = 10^3\,\frac{rad}{s}$

$Z_{C_2} = -j\frac{1}{\omega C_2} = -j\,k\Omega$

Resolvemos, por ejemplo, simplificando en lo posible mediante asociaciones:



$$Z_{eq} = R_2 // Z_{C_2} = \frac{2-4j}{5}\,k\Omega$$

$$I_g = \frac{\bar{e}_g}{R_g + Z_g} = \frac{1}{2}(3+j)\,mA$$

$$V_{C_2} = I_g \cdot Z_{eq} = (1-j)\,V = \sqrt{2}\,e^{-j\frac{\pi}{4}}\,V$$

La expresión temporal $v_{C_2}(t) = \sqrt{2}\cos(10^3 t - \frac{\pi}{4})\,V$

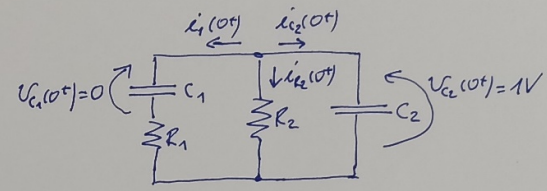
En el instante justo antes de la conmutación (en $t=0^-$)

$$v_{C_2}(0^-) = \sqrt{2}\cdot\frac{1}{\sqrt{2}} = 1\,V$$

Justo después de la conmutación la tensión en el condensador no cambia: $\boxed{v_{C_2}(0^+) = v_C(0^-) = 1\,V}$

P1

b) En $t=0^+$, justo después de la conmutación $\begin{cases} v_{C_2}(0^+) = 1\,V \\ v_{C_1}(0^+) = v_C(0) = 0 \end{cases}$

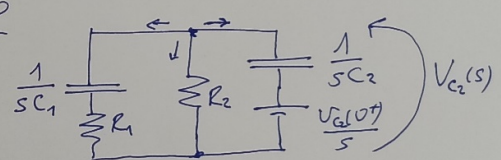


$$i_{R_2}(0^+) = \frac{v_{C_2}(0^+)}{R_2} = \frac{1\,V}{2\,k\Omega} = \frac{1}{2}\,mA$$

$$i_1(0^+) = \frac{v_{C_2}(0^+) - v_{C_1}(0^+)}{R_1} = \frac{1-0}{4} = \frac{1}{4}\,mA$$

$$\boxed{i_{C_2}(0^+) = i_1(0^+) + i_{R_2}(0^+) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}\,mA}$$

c) Para $t \geq 0$



$$\frac{V_{C_2}(s)}{R_1 + \frac{1}{sC_1}} + \frac{V_{C_2}(s)}{R_2} + \frac{V_{C_2}(s) - \frac{v_{C_2}(0^+)}{s}}{\frac{1}{sC_2}} = 0$$

$$\Rightarrow V_{C_2}(s)\left[\frac{1}{R_1 + \frac{1}{sC_1}} + \frac{1}{R_2} + sC_2\right] = \frac{v_{C_2}(0^+)}{s}\,sC_2 \qquad \left| \times(R_1 + \frac{1}{sC_1}) \right.$$

$$\Rightarrow V_{C_2}(s) = \frac{v_{C_2}(0^+)\cdot C_2}{C_2 \cdot s + \frac{1}{R_2} + \frac{1}{R_1 + \frac{1}{sC_1}}} \qquad \left| \times(R_1 + \frac{1}{sC_1}) \right.$$

(Hay que conseguir que en el denominador esté el polin. característico)

P1

$$V_{C_2}(s) = \frac{v_{C_2}(0^+)\cdot C_2\left(R_1 + \frac{1}{sC_1}\right)}{s\cdot C_2\left(R_1+\frac{1}{sC_1}\right) + \frac{1}{R_2}\left(R_1+\frac{1}{sC_1}\right)+1}$$

$$= \frac{v_{C_2}(0^+)\cdot C_2\left(R_1+\frac{1}{s\cdot C_1}\right)}{s\cdot C_2R_1 + \frac{C_2}{C_1} + \frac{R_1}{R_2} + \frac{1}{s\cdot R_2\cdot C_1}+1} \quad \left| \times \frac{s}{C_2R_1}\right.$$

$$= \frac{v_{C_2}(0^+)\left(s+\frac{1}{C_1\cdot R_1}\right)}{s^2 + s\cdot\frac{1}{C_1R_1} + s\frac{1}{C_2\cdot R_2} + \frac{1}{R_1R_2C_1C_2} + s\frac{1}{C_2R_1}}$$

$$\boxed{V_{C_2}(s) = \frac{v_{C_2}(0^+)\left(s+\frac{1}{C_1\cdot R_1}\right)}{s^2 + \underbrace{\left[\frac{1}{C_1R_1}+\frac{1}{C_2R_2}+\frac{1}{C_2R_1}\right]}_{\frac{C_2R_2+C_1R_1+C_1R_2}{C_1C_2R_1R_2}}\cdot s + \frac{1}{R_1R_2C_1C_2}}}$$

d) La ec característica $s^2 + \frac{C_2R_2+C_1R_1+C_1R_2}{C_1C_2R_1R_2}s + \frac{1}{R_1R_2C_1C_2} = 0$

queda: $s^2 + \underbrace{1250}_{2\xi\omega_0}\cdot s + \underbrace{25\cdot10^4}_{\omega_0^2} = 0$

$$\Rightarrow \begin{cases} \omega_0 = \sqrt{25\cdot10^4} = 500 \text{ rad/s} \\ 2\xi\omega_0 = 1250 \end{cases} \Rightarrow \boxed{\xi = \frac{1250}{2\omega_0} = 1{,}25 > 1}$$

Circuito es sobreamortiguado

P1

e) Las raíces del polinomio característico $s^2 + 1250\cdot s + 25\cdot\omega^4 = 0$ resultan: $\begin{cases} s_1 = -250 \\ s_2 = -1000 \end{cases}$

$$\Rightarrow V_{C_2}(s) = \frac{s+500}{(s-s_1)(s-s_2)} = \frac{s+500}{(s+250)(s+1000)}$$

$$\stackrel{!}{=} \frac{A}{s+250} + \frac{B}{s+1000} \quad \text{(desarrollo en fracciones simples)}$$

$$\Rightarrow s+500 = A(s+1000) + B(s+250)$$

para $s=-1000 \Rightarrow -500 = B(-700) \Rightarrow B = \frac{500}{750} = \frac{2}{3}$

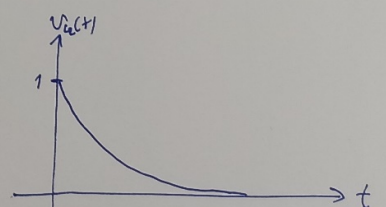
$s=-250 \Rightarrow 250 = A(750) \Rightarrow A = \frac{250}{750} = \frac{1}{3}$

$$\Rightarrow V_{C_2}(s) = \frac{1/3}{s+250} + \frac{2/3}{s+1000}$$

La transformada de Laplace para $t \geq 0$

$$\boxed{v_{C_2}(t) = \left(\frac{1}{3}e^{-250t} + \frac{2}{3}e^{-1000t}\right)u(t)}$$

$v_{C_2}(0) = \frac{1}{3} + \frac{2}{3} = 1$



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P2

$$\begin{cases} I_1 = g_{11}V_1 + g_{12}I_2 \\ V_2 = g_{21}V_1 + g_{22}I_2 \end{cases}$$

$$g_{11} = \left.\frac{I_1}{V_1}\right|_{I_2=0} = \frac{1}{Z_{HN}} = \frac{1+2j}{5}\,\Omega^{-1}$$

$$g_{21} = \left.\frac{V_2}{V_1}\right|_{I_2=0} = \frac{3-j}{3-j} = 1$$

$$V_1 = I_1 \cdot Z_{HN} = \frac{\mathcal{E}_g}{R_g + Z_{HN}} \cdot Z_{HN} = (3-j)\,V$$

Q simétrico $\Rightarrow \begin{cases} g_{12} = -g_{21} = -1 \\ |g| = 1 \Rightarrow g_{11}g_{22} - g_{12}g_{21} = 1 \end{cases}$

$$\Rightarrow g_{22} = \frac{1 + g_{12}g_{21}}{g_{11}} = 0$$

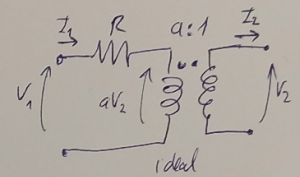
$$\boxed{g = \begin{pmatrix} \frac{1+2j}{5}\,\Omega^{-1} & -1 \\ 1 & 0 \end{pmatrix}}$$

P3

Asociación serie-paralelo de cuadripolos, parámetros "h"

$$\begin{cases} V_1 = h_{11}I_1 + h_{12}V_2 \\ I_2 = h_{21}I_1 + h_{22}V_2 \end{cases}$$

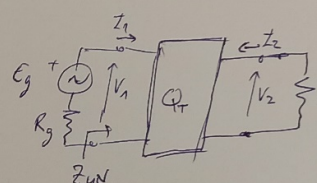
Parámetros "h" del Q de arrastre:



$$\begin{cases} V_1 = RI_1 + aV_2 \\ I_2 = aI_1 \end{cases} \Rightarrow h(Q_{arrast}) = \begin{pmatrix} R & a^{-2} \\ a^{-2} & 0 \end{pmatrix}$$

Asociación de los dos quadripolos (se cumple pues las corrientes de circulación son nulas debido al generador ideal)

$$h_T = h(Q_{arrast}) + h(Q_{adjoint}) = \begin{pmatrix} R+5 & -3+a \\ 3+a & \frac{1}{2}\,\Omega^{-1} \end{pmatrix} = \begin{pmatrix} R+5 & -1 \\ 1 & \frac{1}{2}\,\Omega^{-1} \end{pmatrix}$$



$$V_1 = (R+5)I_1 - 1V_2 \rightarrow Z_{HN} = \frac{V_1}{I_1} = (R+5) - 1\cdot\frac{V_2}{I_1}$$

$$I_2 = I_1 + \frac{1}{2}V_2$$

$$V_2 = 2(-I_2)$$

$$\Rightarrow -\frac{1}{2}V_2 = I_1 + \frac{1}{2}V_2 \Rightarrow \left(-\frac{1}{2} - \frac{1}{2}\right)V_2 = I_1 \Rightarrow \frac{V_2}{I_1} = -1$$

$$\Rightarrow Z_{HN} = R + 5 + 1 \overset{!}{=} R_g$$

$$\Rightarrow \boxed{R = 4\,\Omega}$$